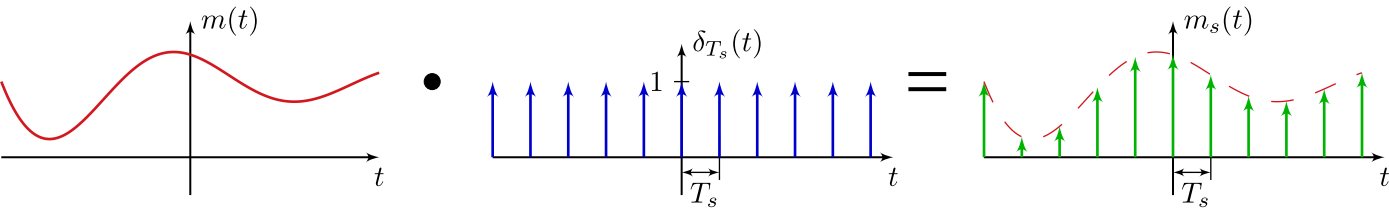


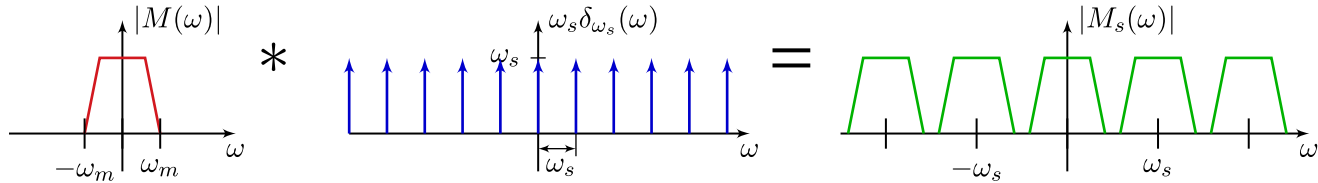
Im Zeitbereich:



$$\begin{aligned} m_s(t) &= m(t) \cdot \delta_{T_s}(t) \\ &= m(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \\ &= \sum_{n=-\infty}^{\infty} m(nT_s) \cdot \delta(t - nT_s) \end{aligned}$$

$$\text{mit } \delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

Im Frequenzbereich:



$$\begin{aligned} M_s(\omega) &= \frac{1}{2\pi} M(\omega) * (\omega_s \cdot \delta_{\omega_s}(\omega)) \\ &= \frac{1}{2\pi} M(\omega) * (\omega_s \cdot \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)) \\ &= \frac{\omega_s}{2\pi} \sum_{n=-\infty}^{\infty} M(\omega) * \delta(\omega - n\omega_s) \\ &= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} M(\omega - n\omega_s) \end{aligned}$$

$$\text{mit } \omega_s \cdot \delta_{\omega_s}(\omega) = \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$